

Ayush Gupta

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Summary

Software Engineer focused on backend architecture, Machine Learning infrastructure, and distributed systems. Active open-source contributor to Kubeflow. Experienced in optimizing system latency, building full-stack applications, and writing production-ready, heavily tested code.

Experience

Mar 2026 - PRESENT	● Open Source Contributor Kubeflow - Katib (AutoML): Fixed HuggingFace trainer startup crashes by implementing a robust JSON parser for training_parameters. Ensured backward compatibility for empty payloads and added comprehensive pytest coverage for edge cases (PR #2656). - Docs-Agent: Contributed to the documentation agent's architecture, optimizing backend retrieval and data handling to improve response reliability (PR #186).
Jan 2026 - PRESENT	● Contributor DSC Winter of Code Contributed to open-source repositories, focusing on bug fixes, code maintainability, and passing automated CI/CD checks during technical reviews

Education

2024 - 2028	● VIT Bhopal University, Bhopal B.Tech In Computer Science & Engineering CGPA - 7.61
2022 - 2023	● Brij Kunwar Devi Aldrich Public School (CBSE) Class XII Percentage - 74%

Skills

• Data Structures & Algorithms	• SQL	• CI/CD Pipeline	• Kubeflow	• HuggingFace
• Object-Oriented Programming (OOP)	• JavaScript	• Pytest	• TensorFlow	
• System Architecture	• Java	• Next.js	• PyTorch	
• Python	• Relational Databases	• React	• OpenCV	
• C++	• GitHub Actions	• Node.js	• Pandas	

Projects

AI Surveillance System | Python, CNN, RNN, OpenCV

📅 Mar 2026 – Apr 2026

🔗 [Project Link](#)

- Engineered a deep learning pipeline combining spatial (CNN) and temporal (RNN) video analysis to achieve real-time anomaly detection in CCTV streams.
- Designed a parallel streaming and inference pipeline using Python multiprocessing, ensuring low-latency video processing for continuous surveillance.
- Implemented computer vision models for automated activity recognition, significantly reducing the manual effort required for threat identification.

HerWay – Predictive Safe Route Navigation System | Next.js, Machine Learning, Geospatial APIs

📅 Jan 2026 – Apr 2026

🔗 [Project Link](#)

- Built an AI-driven safety navigation system that identifies safer travel routes by analyzing real-time factors like location data, crime rates, and proximity to emergency services using machine learning models.
- Developed real-time tracking and emergency response features, enabling SOS alerts and live location sharing to improve user safety during travel scenarios.
- Designed and integrated a scalable mobile architecture (Android + backend services) with location-based APIs and data pipelines to support continuous route evaluation and user interaction.

Sunrise Hotel Website | Next.js, TypeScript, Tailwind CSS, Supabase

📅 Aug 2025 – Nov 2025

🔗 [Project Link](#)

- Built a production-ready, full-stack hotel booking platform using Next.js (TypeScript) and Tailwind CSS, optimizing for fast load times and responsive UX.
- Architected a secure backend using Supabase, implementing SMTP-based email authentication to improve user management and reliability.
- Deployed the application via Vercel with modern CI/CD workflows, ensuring high availability, consistent uptime, and seamless updates.

Yoga Pose Detection Engine | Python, OpenCV, MediaPipe, NumPy

📅 Jul 2025 – Nov 2025

🔗 [Project Link](#)

- Implemented computer vision-based keypoint detection models to classify human poses in real-time, greatly improving posture accuracy detection.
- Mapped dynamic body joint angles to predefined pose templates to deliver automated, instant feedback for user posture correction.
- Optimized the video frame processing loop and integrated lightweight pose models to guarantee efficient inference and a smooth user experience.

Achievements

Open Source Architect - Kubeflow

Successfully merged core architecture and bug-fix pull requests into Kubeflow (Katib and Docs-Agent), directly impacting a globally utilized machine learning infrastructure platform

DSC Winter of Code 2026

Selected from a highly competitive applicant pool as an open-source contributor to maintain and scale community-driven repositories.

#T20DSACHallenge Competitor

Completed an intensive data structures and algorithms challenge, demonstrating advanced proficiency in Stacks, Queues, and complex algorithmic problem-solving.

Technical Certifications

Applied Machine Learning in Python (University of Michigan / Coursera) & *Introduction to Generative AI* (Google Cloud Skills Boost).

Responsibility

MATRIX Club, VIT Bhopal University

Core Member - Technical Team

Collaborate with team members to organize coding competitions and technical workshops. Mentor junior students in Python programming and Data Structures.

TechnoMech Club, VIT Bhopal University

Core Member - Technical Team

Architected and deployed internal campus tools for 5,000+ users as a Core Technical Member, improving system load times by 82% and establishing strict version control practices.